



11. INTRODUCTION TO CHECKING COMPONENTS

MIKROELEKTRONIKA

So you've put a circuit together and as far as you know everything appears to be ok, but it doesn't work as expected. Even worse, it refuses to give any signs of life. What do you do? First, check the circuit for mechanical failures, like non-connected wires, broken vias on the board (these are holes on the printed circuit board that have a metal coating down the length of the hole to connect one side of the board to the other), bad battery contacts inside the case, broken pins on a component, cold solder joints, etc.

If this doesn't come up with a result, you should compare values of components with the schematic.

You may have put a component in the wrong place, or read values the wrong way. Maybe you forgot k in front of Ohms. Maybe you connected the supply to the wrong pin of an IC.

The next step is to test each component on the board.

Start troubleshooting by measuring DC voltages at certain points of the board, and comparing these values to the schematic. So, by knowing the operation of the circuit you start the process of elimination to find the "suspect" component.

If there are several "suspects", and this is not a rare occurrence in complex devices, the testing is divided into groups of components. You start checking in reverse soldering order, this means you start with components last soldered, because those are the most sensitive components on the circuit like integrated circuits, transistors, diodes, etc.

The fastest and simplest method to troubleshoot is to use an "ohm-meter."

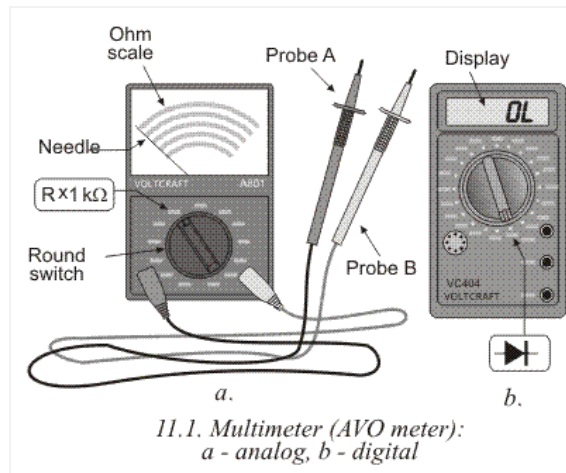
In most cases you don't have an ohm-meter by itself as it is usually added to an ammeter and voltmeter in one instrument, called AVO meter or multimeter.

The safest and most accurate method is to desolder the component from the board when testing it, because other components could lead to a wrong diagnosis, so you have to be very careful when testing in-circuit.

Ok, you should know something about multimeters now. There are two kinds: analog and digital. Analog ones are items of the past, and since they use a needle to tell you values, it can be difficult determining the right value.

Digital meters, on the other hand have a display. You should go for this type, although both come in different sizes and with different ranges. Their price is from several dollars, to several hundreds of dollars for really good professional types.

Two instruments are shown in 11.1.



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